

Russell Dudek

APPLIED AI PRODUCT & OPERATING SYSTEMS

Operator-technologist who turns emerging AI into adopted operating capability. Builds the mechanisms between product intent and real work: workflow discovery, structured context, agentic patterns, human authority, evaluation, governance, adoption, and measurable outcomes.

Role-Aligned Experience

- Leads AI-first operating transformation across production, purchasing, quality, customer success, support, engineering issue flow, ERP/Odoo, Jira, HubSpot, Slack, telemetry, and AI-assisted workflows.
- Connects factory, field, customer, and machine-fleet signals into prioritized work, earlier risk detection, cleaner handoffs, dashboards, standard work, training, and accountability.
- Works directly in a software-enabled robotics and machine-vision environment with approximately 1,000 fielded robots informing support and operating feedback loops.
- Develops AI-readiness and opportunity frameworks that assess value, feasibility, data readiness, governance, adoption burden, reuse, and time-to-value.
- Designs agentic workflow patterns using structured context, retrieval, AI agents, human judgment, escalation, decision records, evaluation, and governance rails.
- Advises on AI-enabled operations, logistics technology, warehouse automation, AMRs, ASRS, cobots, transportation systems, network analysis, and workflow design.
- Led launch and execution for Amazon Prime Pittsburgh in a 24/7 customer-backward environment supporting approximately five million second-day orders annually.
- Built leadership routines, labor planning, throughput visibility, escalation paths, standard work, and operating mechanisms.
- Applied Lean and Gemba practices across safety, quality, throughput, customer experience, and team engagement; contributed to Prime Air research and Amazon Flex geolocation concepts.

Applied AI Product Operating Model

Frame the work

Define the user, recurring decision, client outcome, baseline, and friction before selecting technology.

Bound the system

Make data, tools, human authority, evaluation, escalation, ownership, and stop conditions explicit.

Embed adoption

Design the workflow, behavior change, enablement, support, and operating cadence as part of the product.

Prove and compound

Instrument quality, use, economics, and outcomes; harvest reusable assets and retire weak ideas.

Russell Dudek

APPLIED AI PRODUCT & OPERATING SYSTEMS

Additional Experience

- Progressed through R&D program management, production management, strategic acquisition, operations management, and director-level leadership.
- Led engineering, manufacturing, quality, customer delivery, and complex technical programs in advanced electronics and high-reliability environments.
- Built quality systems, standard work, root-cause practices, performance measures, and process controls.
- Collaborated with high-consequence customers and partners including DoD, CERN, Intel, and AMD; published industry articles, delivered technical talks, and participated in the HyperTransport Consortium.
- Built the company's first remote regional operation.
- Modernized distributed customer, CRM, commercial, fulfillment, logistics, digital marketing, SEO, and online-buying workflows.
- Advanced AI-enhanced transportation-management and route-planning concepts.
- Led regulated drone operations across medical, energy, logistics, real-estate, and restricted-airspace contexts with a 100% safety record and no FAA incidents.
- Developed TensorFlow-based concepts for automated aerial inventory and computer-vision-enabled field intelligence.
- Designed medical drone delivery and chain-of-custody concepts connecting technology, risk, stakeholders, and governed execution.
- Co-founded and led a public-private impact initiative focused on sustainable infrastructure, green finance, and civic transformation.
- Helped shape and evaluate a \$200 million Social Impact Bond feasibility concept for Pittsburgh sewage and green-infrastructure improvement.

Product Posture

Start with the recurring workflow and decision. Define the client outcome, human authority, trust boundary, reusable component, owner, and evidence before selecting the AI implementation.

Technical Toolkit

Agentic AI | LLM workflows | RAG patterns | Human-in-the-loop design | Python | SQL | JavaScript | TensorFlow | Tableau | ERP/Odoo | Jira | Confluence | HubSpot | Slack | GIS and mapping

Education

University of Pittsburgh Bachelor of Science Academic grounding includes materials science, physical chemistry, and biology-related studies.

Google Scholar profile

Credentials

Google AI Essentials IBM Enterprise Design Thinking Practitioner Six Sigma Certification OSHA 10

Leadership & Community

IEEE | CMPT and EDS Pittsburgh Chapter Chair, 2012-2016. Led regional technology programming connecting industry, academia, STEM outreach, electronics, manufacturing, and emerging technology.

Candidate Thesis

Make AP's best client work compound through governed, reusable products that preserve expert judgment and earn adoption.

Selected Operating Patterns

Signal to decision

Connect fragmented operating, customer, quality, and technical signals into prioritized work, ownership, and escalation.

Evidence over enthusiasm

Make expert judgment and human authority explicit; use verified outcomes or named measurement categories, never inflated deployment claims.